**Keywords:** omni-modal benchmarking • multimodal LLMs • ambiguity resolution • audio-visual integration

# TRIAD: Benchmarking Omni-Modal Ambiguity in Multimodal Large Language Models

*A New Benchmark Reveals That Even the Best Omni-Modal Systems Trail Humans by Over 20 Points When All Three Modalities Are Needed to Resolve a Scene*

**By Zhaolu Kang, Yidi Wang, Yile Li, Siqi Zeng, Yingjie He, Richeng Xuan, Zhichao Hu**

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Multimodal language models that process text, images, and audio are routinely evaluated on benchmarks where a single dominant modality carries the answer—leaving intact an unmeasured gap between claimed omni-modal capability and the genuine ability to fuse all three modalities simultaneously. TRIAD closes this gap with 327 hand-crafted queries spanning 86 scene groups and 18 ambiguity categories, each designed so that the complete text–image–audio triplet yields a unique answer while any single-modality subset leaves two or more options open. Across 14 state-of-the-art omni-modal systems, the best model trails humans by over 20 group-accuracy points, and removing answer options drives model group accuracy to single digits.

**AT A GLANCE**

**Problem:** Existing benchmarks do not require models to integrate all three modalities; a single modality usually suffices to answer correctly.

**Why it matters:** Omni-modal systems are deployed as universal assistants; a model that silently ignores audio will fail in safety-critical settings where acoustic context is essential.

**Method:** 327 text-image-audio triplets where the correct answer requires all three modalities, evaluated on item and scene-group accuracy with and without answer options.

**Result:** Best model trails humans by >20 points on group accuracy; removing answer options drops model group accuracy to single digits.

**Evidence:** 14 state-of-the-art omni-modal systems on 86 scene groups covering 18 ambiguity categories.

## THE BIG PICTURE

The ambition of omni-modal language models—processing text, image, and audio within a single system—promises agents that perceive the world as richly as humans do. Yet evaluating this ambition has relied on benchmarks that are functionally bimodal: image-question answering where text alone is sufficient, or audio tasks where the image adds marginal signal.

TRIAD operationalizes modal indispensability: for each query, every modality is necessary and no proper subset is sufficient. The dataset captures everyday scenes where text, image, and audio together resolve what neither pair can—e.g., hearing an alarm while seeing a clock while reading a date.

## WHY EXISTING APPROACHES FALL SHORT

**Item-level evaluation is misleading.** A model achieving 70% item accuracy may still fail every scene group, because group accuracy requires consistent disambiguation across all items sharing a scene.

**Answer options inflate performance.** Multiple-choice formats let models exploit statistical regularities in the option set. Performance drops to single digits in the open-ended condition, confirming option exploitation.

**Audio cues are underweighted.** Models default to text–image reasoning even when the disambiguating signal is acoustic, exposing a systematic failure mode in current architectures.

## CORE MECHANISM

TRIAD's benchmark construction follows three stages: (1) Scene collection: authors collect or synthesize naturalistic scenes representable in all three modalities with complementary but partially overlapping information. (2) Ambiguity verification: each triplet is manually reviewed to confirm that any two-modality combination leaves at least two consistent answer options. (3) Query authoring: questions are written to be answerable only with the full triplet; distractors are crafted to be plausible given incomplete modality access.

86 scene groups share a common real-world context and each contain 3–5 queries, allowing the group accuracy metric to test whether disambiguation generalizes within a scenario.

## TECHNICAL FORMULATION

Let T = (t, v, a) denote a triplet of text t, image v, and audio a, and let A = {a_1, ..., a_K} be the answer set. A triplet is modally indispensable if:

argmax_k P(a_k | T) is unique, but

argmax_k P(a_k | T \ {m}) has |>1 elements for every modality m in {t, v, a}.

Group accuracy for scene group g is: GroupAcc(g) = 1[forall q in g, a_hat(q) = a*(q)].

Mean group accuracy across all 86 groups is the primary metric.

## LEARNING OR INFERENCE PROCEDURE

TRIAD is a static evaluation benchmark; no training is involved. Models are prompted with the triplet in their native input format (text + image + audio), asked to output either a multiple-choice letter or a free-form answer, and scored automatically. Human performance is estimated from a held-out annotator pool distinct from the construction team.

## WHAT THE GUARANTEE SAYS

**Soundness:** A model that answers correctly must have accessed all three modalities (no single-modality shortcut exists by construction).

**Completeness:** Group accuracy measures consistent disambiguation within a scene, ruling out lucky individual item guesses. These are guarantees about the benchmark’s diagnostic validity, not about any particular model.

## EXPERIMENTAL FINDINGS

**Group accuracy gap:** The best of 14 evaluated omni-modal systems achieves group accuracy >20 points below human performance.

**Item vs. group accuracy:** Models with high item accuracy (e.g., >65%) still fail most scene groups, confirming inconsistency.

**Open-ended collapse:** Removing answer options drops average model group accuracy to single digits; human group accuracy declines only modestly.

**Audio underuse:** On audio-indispensable items, models score near chance even in the multiple-choice condition.

## ABLATIONS AND INTERPRETATION

**Modality ablation:** Providing models with only text+image, text+audio, or image+audio gives performance close to random, confirming benchmark validity.

**Model family:** Closed-source frontier models outperform open-source models by a larger margin on TRIAD than on existing benchmarks.

**Category breakdown:** The hardest categories involve temporal audio events and spatial sound-source localization.

## REFERENCE

Zhaolu Kang, Yidi Wang, Yile Li, Siqi Zeng, Yingjie He, Richeng Xuan, Zhichao Hu. **TRIAD: Benchmarking Omni-Modal Ambiguity in Multimodal Large Language Models.** arXiv:2610.30841. NeurIPS 2026. https://arxiv.org/abs/2610.30841

**Keywords:** multi-objective RL • ensemble policies • trajectory attribution • objective conflict

# Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution

*Objective-Conflict Coefficients Quantify Preference Trade-offs, While Routed Ensembles and Kernel-Surrogate Attribution Close the Multi-Objective Efficiency Gap*

**By Zhenshuo Zhang, Ruoxuan Xiong, Hongyang R. Zhang**

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Multi-objective reinforcement learning demands a single agent to simultaneously satisfy multiple reward objectives weighted by a user-supplied preference vector—a task made hard when objectives conflict and gradient directions oppose each other. This paper introduces objective-conflict coefficients that formally quantify this conflict, derives matching online regret bounds, and proposes two practical algorithms: a routed ensemble that routes each trajectory to the basis policy with highest expected return under its preference weight, and a trajectory attribution module based on kernel ridge regression that identifies which trajectories are most valuable for which preference objectives.

**AT A GLANCE**

**Problem:** A single preference-conditioned policy fails when objectives strongly conflict; the degree of conflict was not previously formalised.

**Why it matters:** MORL is central to robotics, resource allocation, and fairness-aware learning, where trade-offs among objectives must be navigated at deployment.

**Method:** Objective-conflict coefficients + routed ensemble policy learning (trajectory routing to K basis policies) + kernel ridge regression trajectory attribution.

**Result:** Consistent improvements over single-policy and standard ensemble baselines on multi-objective robotic control benchmarks.

**Evidence:** Multi-objective MuJoCo locomotion benchmarks; formal regret analysis as a function of maximum conflict coefficient.

## THE BIG PICTURE

In multi-objective RL (MORL), the agent receives a vector reward r = (r_1,...,r_K) and a preference weight w in the simplex. The goal is to maximize w^T J(pi) across many possible w, learning the entire Pareto front of preference-weighted returns.

Conflict arises when grad J_i and grad J_j point in opposing directions, forcing any gradient step to damage at least one objective. Prior approaches either concatenate w to the state (ignoring conflict) or use hand-crafted ensemble partitions (ignoring data efficiency). This work addresses both gaps.

## WHY EXISTING APPROACHES FALL SHORT

**Conditioned single policy:** Appending w to the state does not prevent gradient conflicts. For strongly conflicting objectives, the single policy converges to a suboptimal compromise rather than the Pareto front.

**Naive ensembles:** Manually partitioning the preference space into K fixed regions ignores that trajectories generated under one preference weight may be highly informative for another.

**No formal conflict measure:** Without a quantitative measure, there is no principled way to predict how many basis policies are needed.

## CORE MECHANISM

Routed Ensemble Policy Learning maintains K basis policies {pi_1,...,pi_K} and iterates: (1) Collect trajectories by sampling basis policies under current preference weights. (2) Route: assign each trajectory tau_j to basis policy k* = argmax_k w_j^T J_k(pi_k). (3) Update: apply PPO to each pi_k using only routed trajectories.

Trajectory Attribution uses a kernel ridge regression surrogate f_hat(pi, w; D) to estimate policy performance as a function of the training trajectory set D, enabling curriculum sampling of high-value trajectories.

## TECHNICAL FORMULATION

Objective-Conflict Coefficient for objectives i and j:

c_ij(pi) = -grad_theta J_i . grad_theta J_j / (||grad J_i|| ||grad J_j||)

Regret bound: Regret(T) = O(sqrt(T * c_max)) where c_max is the max conflict.

Kernel surrogate: f_hat(w; D) = y^T (K + lambda I)^{-1} k(w)

where K_ij = kappa(phi(tau_i), phi(tau_j)) and y_j = w_j^T J(pi).

## LEARNING OR INFERENCE PROCEDURE

Each training step: (1) Sample preference weights from the prior. (2) Collect trajectories by rolling out each basis policy. (3) Compute attribution scores; upsample high-attribution trajectories. (4) Route trajectories to basis policies and apply PPO updates.

At test time, given preference w*, evaluate J_k(pi_k) for each basis policy and return the best.

## WHAT THE GUARANTEE SAYS

The regret bound O(sqrt(T * c_max)) provides an instance-dependent characterisation: MORL is hard when objectives conflict strongly and easy when they are compatible. The routed ensemble achieves this rate for any K >= c_max / epsilon. This is the first formal characterisation of MORL hardness in terms of objective geometry.

## EXPERIMENTAL FINDINGS

**Multi-objective MuJoCo:** Routed ensemble outperforms conditioned single-policy baselines by 8–15% in hypervolume metric across 4 multi-objective task suites.

**Vs. naive ensemble:** Attribution-based curriculum sampling improves sample efficiency by ~30% (same hypervolume with fewer environment interactions).

**Conflict coefficient:** Empirical c_max correlates strongly with the observed performance gap (r = 0.81), validating the theoretical predictor.

**Scalability:** Performance improvements hold for K in {2, 4, 8} basis policies; gains plateau at K = 8.

## ABLATIONS AND INTERPRETATION

**Routing vs. random assignment:** Random assignment reduces hypervolume by 7%, confirming conflict-aware routing is the key driver.

**Attribution vs. uniform sampling:** Replacing kernel attribution with uniform sampling costs ~20% in sample efficiency.

**Conflict coefficient threshold:** Task suites with c_max < 0.1 show no significant gain from ensembles—a practical criterion for when ensembles are needed.

## REFERENCE

Zhenshuo Zhang, Ruoxuan Xiong, Hongyang R. Zhang. **Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution.** arXiv:2610.35733. NeurIPS 2026. https://arxiv.org/abs/2610.35733

**Keywords:** video diffusion • novel view synthesis • autonomous driving simulation • geometry grounding

# GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation

*LiDAR-Warped Reference Frames and Synthetic Para4D Data Enable Zero-Shot Lane-Shift Video Generation on Real-World Waymo and nuScenes Benchmarks*

**By Yuchen Xi, Yuhan Guo, Chenwei Hou, Zixu Liu, Jin Fang, Jason Liu, Ruigang Yang**

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Autonomous driving simulators require realistic video from camera poses that were never recorded—a lane shift, a slightly different heading, or a hypothetical trailing vehicle. GeoV2V addresses this challenge by combining LiDAR and depth-based geometric priors with full-video conditioning on the Wan 2.1 video diffusion backbone, generating temporally consistent, geometrically accurate driving videos under large trajectory shifts. Trained entirely on Para4D, a new synthetic CARLA-based dataset of synchronized multi-lane videos, GeoV2V transfers zero-shot to real-world Waymo and nuScenes data, achieving state-of-the-art visual quality and geometric consistency.

**AT A GLANCE**

**Problem:** Large viewpoint shifts in novel-view driving video reveal occluded regions; pure generation loses geometric coherence across frames.

**Why it matters:** High-fidelity novel-view synthesis is essential for scalable simulation of rare events (lane changes, near-misses) that cannot be captured by fixed sensor rigs.

**Method:** Geometric warping via LiDAR/depth priors + Wan 2.1 video diffusion conditioned on warped sequence + Para4D synthetic training data.

**Result:** State-of-the-art FID, FVD, and depth-consistency on Waymo and nuScenes without fine-tuning on real data.

**Evidence:** Waymo, nuScenes, and Para4D benchmarks; ablations isolating geometric conditioning and synthetic pretraining.

## THE BIG PICTURE

The long tail of driving scenarios—rare lane changes, unexpected obstacles, adverse weather—cannot be covered by fixed-route data collection. Novel view synthesis (NVS) addresses this by generating plausible videos from camera poses never recorded.

GeoV2V grounds a powerful video diffusion prior (Wan 2.1) with explicit 3D geometry, turning the generation problem into geometry-aware inpainting: warp what is visible, inpaint what is occluded, and preserve temporal dynamics.

## WHY EXISTING APPROACHES FALL SHORT

**Image-based NVS (NeRF, 3DGS):** Require dense multi-view captures unavailable in single-camera drive-by settings.

**Pure video generation (Sora, Gen-3):** Unconstrained generation ignores 3D structure, producing temporally inconsistent lane layouts.

**Warping-only methods:** Geometric warp leaves holes where occluded content is revealed; filling them with inpainting lacks temporal coherence.

## CORE MECHANISM

GeoV2V operates in three stages: (1) Geometric warping: LiDAR point clouds project each reference frame to the target pose, producing a warped reference sequence with explicit holes where geometry is occluded. (2) Conditional video generation: Wan 2.1 video diffusion transformer takes the warped sequence as conditioning via cross-attention, filling holes and refining artifacts. (3) Para4D training: The model trains on synthetic CARLA multi-lane videos covering diverse weather, lighting, and traffic conditions.

## TECHNICAL FORMULATION

Warp operator W using depth D_t and camera intrinsics K:

I_tilde_t = W(I_t^ref, D_t, P_t^ref -> P_t^tgt, K)

Video diffusion model G_theta generates:

V_out = G_theta(z_T, V_tilde, M)

where z_T is Gaussian noise and M are per-pixel visibility masks.

Training objective: L = E[||G_theta(z_t, V_tilde, M) - V_gt||^2] on Para4D pairs.

## LEARNING OR INFERENCE PROCEDURE

**Training:** GeoV2V trains entirely on synthetic Para4D data with lateral offsets of 1.0–5.0 m, covering standard and extreme viewpoints. Wan 2.1 weights are initialised from a pretrained checkpoint.

**Inference (zero-shot):** On Waymo and nuScenes, only LiDAR-based depth is used for warping; the generator runs without any domain-specific fine-tuning.

## WHAT THE GUARANTEE SAYS

The geometric warping step provides an explicit constraint: visible pixels in the reference frame are reproduced at the correct target location with sub-pixel registration accuracy (verified by LiDAR reprojection error). The diffusion model only generates content in provably occluded regions (mask M_t = 0), giving a formal separation between geometric reliability and generative uncertainty.

## EXPERIMENTAL FINDINGS

**Waymo lane-shift NVS:** State-of-the-art FID and FVD against prior methods (WoVoGen, GeoFlow, AnyScene), with >15% FVD improvement at 3.5 m lane shifts.

**nuScenes:** Consistent gains in geometric depth consistency (homography error reduction >20%) and visual quality metrics.

**Challenging lighting:** GeoV2V maintains FVD gains in night and adverse weather where geometric priors are noisiest.

**Para4D ablation:** Replacing Para4D with a warping-only baseline increases FVD by 38%, confirming the importance of explicit cross-lane supervision.

## ABLATIONS AND INTERPRETATION

**LiDAR vs. monocular depth:** LiDAR improves depth consistency by 12%; monocular depth is sufficient for visual quality but not geometric accuracy.

**Conditioning mechanism:** Replacing cross-attention with concatenation reduces FVD by 8%, confirming that temporal context is best exploited through attention.

**Training data scale:** 10k Para4D pairs is sufficient; doubling to 20k improves FVD by only 2%.

## REFERENCE

Yuchen Xi, Yuhan Guo, Chenwei Hou, Zixu Liu, Jin Fang, Jason Liu, Ruigang Yang. **GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation.** arXiv:2610.25245. ECCV 2026. https://arxiv.org/abs/2610.25245

**Keywords:** pixel diffusion transformers • prediction parameterization • MFlow adaptive interpolation • ConvRefiner

# Rethinking the Design Space of Vanilla Pixel Diffusion Transformers

*MFlow Learns a Per-Direction Interpolation Between x0- and Velocity Prediction, Closing the Quality Gap Between Pixel and Latent Diffusion*

**By Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra**

arXiv:2610.23854 • Under review, 2026

Pixel-space diffusion transformers avoid VAE tokenization artifacts but have lagged behind latent diffusion in generation quality, partly because the optimal prediction parameterization—clean image (x0) versus velocity (u)—depends non-trivially on model capacity and data statistics. MFlow addresses this by introducing per-principal-direction gains that interpolate between x0- and u-prediction, learned end-to-end with diffusion training. Experiments on ImageNet, UCF-101 video, and large-scale text-to-image generation confirm that MFlow consistently outperforms fixed-parameterization baselines, and that larger models systematically prefer velocity prediction. A lightweight ConvRefiner trained post-hoc with an adversarial loss further closes the persistent high-frequency gap.

**AT A GLANCE**

**Problem:** The best prediction parameterization for pixel DiTs varies with model capacity and data; the fixed-choice approach leaves quality on the table.

**Why it matters:** Pixel DiTs promise simpler, higher-fidelity generation than latent diffusion, but their quality gap has blocked adoption.

**Method:** Per-direction gain m_i in [0,1] interpolates x0- and u-prediction; gains trained jointly with the diffusion model. ConvRefiner handles residual high-frequency deficits post-hoc.

**Result:** Competitive or improved FID/FVD vs. fixed-parameterization baselines on ImageNet, UCF-101, and text-to-image.

**Evidence:** Ablations across model scales, architectures, and modalities; systematic m_i analysis validating the capacity-preference relationship.

## THE BIG PICTURE

Latent diffusion models (LDMs) such as SD3 and FLUX compress images into a low-dimensional latent code before diffusion, enabling fast generation but introducing VAE reconstruction artifacts. Pixel diffusion transformers (pixel DiTs) sidestep this by diffusing directly in pixel space, preserving sharp boundaries.

The quality gap between pixel DiTs and LDMs has been attributed to less stable training. This paper identifies an underappreciated cause: the prediction parameterization. The choice between predicting the clean image (x0), the noise (epsilon), or the velocity (u = alpha*x0 - sigma*epsilon) determines the gradient signal across noise levels in ways not studied systematically for pixel DiTs.

## WHY EXISTING APPROACHES FALL SHORT

**Fixed x0-prediction (JiT):** Showed x0-prediction outperforms epsilon-prediction at moderate scale, but this advantage diminishes or reverses at larger model sizes.

**Fixed u-prediction (flow matching):** Works well for latent diffusion but underperforms x0-prediction at smaller pixel DiT scales.

**AsymFlow:** A binary rank-asymmetric assignment is a special case of MFlow with step-function gains, losing the benefit of smooth interpolation.

## CORE MECHANISM

MFlow parameterizes the prediction target for the i-th principal direction as:

y_hat_i = m_i * u_i + (1 - m_i) * (x0)_i

where the scalar gain m_i in [0,1] is the sole additional parameter per direction. m_i = 0 recovers clean-image prediction; m_i = 1 recovers velocity prediction. Gains are initialized from signal-to-noise statistics and updated with gradient descent.

ConvRefiner is a lightweight CNN trained to predict the residual in the frequency domain with a spectral adversarial loss, with the diffusion backbone frozen.

## TECHNICAL FORMULATION

Flow matching diffusion: x_t = alpha_t * x0 + sigma_t * epsilon.

Velocity target: u = alpha_t * x0 - sigma_t * epsilon.

MFlow model f_theta predicts:

y_hat_i(x_t, t) = m_i * f_theta^(u)(x_t, t)_i + (1-m_i) * f_theta^(x0)(x_t, t)_i

Training loss: L_MFlow = E_{t,x0,eps}[sum_i ||y_hat_i - y_i^target||^2]

Gradients flow through both f_theta and {m_i}.

Optimal gain under linear model: m_i* = Var[u_i] / (Var[u_i] + Var[(x0)_i])

## LEARNING OR INFERENCE PROCEDURE

**Training:** Standard diffusion training with an extra gradient term for {m_i}. Gains initialized as m_i = SNR_i / (1 + SNR_i). ConvRefiner trained in a second stage with backbone frozen.

**Inference:** Identical to standard flow-matching sampling (DDIM, DPM-Solver, or Euler). ConvRefiner adds one forward pass with the lightweight CNN.

## WHAT THE GUARANTEE SAYS

MFlow is a strict generalization of fixed x0- and u-prediction: setting all m_i = 0 recovers the x0 loss; all m_i = 1 recovers velocity loss. Therefore MFlow can only improve on either fixed choice—its optima weakly dominate.

The theoretical optimal m_i* is an increasing function of intrinsic dimensionality, explaining why larger models (which represent more directions) benefit more from velocity prediction.

## EXPERIMENTAL FINDINGS

**ImageNet 256x256:** MFlow reduces FID by 15–20% over fixed x0- and u-prediction baselines at the same model scale.

**UCF-101 video:** Consistent FVD improvement; gains are largest for the largest model (+18% over velocity-only baseline).

**Large-scale text-to-image:** MFlow at 1B parameters matches or exceeds prior best pixel DiT at 3B parameters.

**ConvRefiner:** Adds 1.2 ms per image; reduces high-frequency FID deficit by 40% without changing global FID.

**Capacity scaling:** Empirical m_i values shift from near-zero to near-one as model size increases, confirming the theoretical prediction.

## ABLATIONS AND INTERPRETATION

**Gain initialization:** Random initialization converges to similar final values but requires 2x more training steps; SNR-based initialization strongly recommended.

**Per-direction vs. global gain:** A single shared gain m reduces FID by only 60% of the per-direction improvement.

**ConvRefiner training signal:** Replacing spectral GAN with pixel-space L2 loss fails to recover high-frequency detail—adversarial objective is essential.

## REFERENCE

Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra. **Rethinking the Design Space of Vanilla Pixel Diffusion Transformers.** arXiv:2610.23854. Under review, 2026. https://arxiv.org/abs/2610.23854

**Keywords:** test-time adaptation • robot locomotion • reinforcement learning • distribution shift

# TEMPO: Efficient Test-Time Adaptation For Robot Policies

*Critic Entropy Regularization and Score-Based Memory Enable On-Device Policy Adaptation—Up to 80% Performance Recovery on Physical Quadrupeds Under Load*

**By Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos**

arXiv:2610.20318 • NeurIPS 2026

Reinforcement-learned locomotion policies trained in simulation degrade in deployment when conditions shift—payload changes the center of mass, terrain modifies effective friction, sensors drift with temperature. TEMPO addresses this with a surprisingly effective combination: an entropy regularizer on the critic that prevents value collapse under distribution shift, paired with a score-based memory buffer that upweights hard observations during on-device adaptation. Across three quadruped platforms (Go1, Go2, G1) and four shift types in MuJoCo Playground, TEMPO yields up to 80% improvement in the single-device setting; on a physical Unitree Go2 carrying an added payload, TEMPO recovers pre-shift performance within minutes.

**AT A GLANCE**

**Problem:** Sim-to-real locomotion policies degrade when deployment conditions (payload, terrain, sensors) differ from training—and no simulator is available at test time.

**Why it matters:** Real-world deployment requires policies that adapt autonomously; simulator-based fine-tuning is impractical on hardware.

**Method:** Critic entropy regularizer prevents value collapse; score-based memory oversamples observations the policy finds hard.

**Result:** Up to 80% performance improvement in simulation; physical Unitree Go2 recovers from payload shift within 10 minutes.

**Evidence:** Go1, Go2, G1 in MuJoCo Playground under four distribution shifts; physical Unitree Go2 experiment.

## THE BIG PICTURE

Reinforcement learning in simulation remains the dominant approach for robot locomotion, but the sim-to-real gap means deployed policies face conditions their training never covered. Once deployed, the robot's own experience is the only data available for further improvement.

Test-time adaptation (TTA) of neural networks has been studied for supervised classification, but RL policies present a harder challenge: no ground-truth labels, expensive reward evaluation, critic divergence under distribution shift, and compute constrained to a single embedded device.

## WHY EXISTING APPROACHES FALL SHORT

**Standard PPO fine-tuning at test time:** Without a simulator, PPO must use on-device rollouts directly. Distribution shift causes the critic to diverge, destabilising policy updates.

**Domain randomisation:** Broad randomisation reduces peak performance and still fails for shifts outside the randomised range.

**Meta-learning (MAML, PEARL):** Requires a distribution over adaptation tasks at training time—unavailable when the real-world distribution is unknown—and inner-loop gradient computation expensive on embedded hardware.

## CORE MECHANISM

**1. Critic Entropy Regularizer.** The critic loss is augmented with an entropy penalty: L_V = L_TD + beta * H(V_phi(s)), where H(V_phi(s)) is the entropy of the value distribution across a replay batch. This prevents the critic from collapsing to a near-constant value function under distribution shift.

**2. Score-Based Memory.** Each observation s in the replay buffer is assigned a score proportional to its TD error: score(s) proportional to |r + gamma*V(s') - V(s)|. High-scoring observations (where the critic is most wrong) are sampled more frequently, biasing updates toward the hardest parts of the new distribution.

## TECHNICAL FORMULATION

Full TEMPO objective at each adaptation step:

L_TEMPO = L_PPO + lambda_V [L_TD + beta * H(V_phi)]

Score-based sampling probability for transition i:

p_i = score(s_i)^alpha / sum_j score(s_j)^alpha, alpha = 0.6

(Priority exponent follows the PER convention.)

Entropy regularizer guarantee: H(V_phi(s)) >= H_min > 0 for all t, ensuring policy gradient variance stays non-zero throughout adaptation.

## LEARNING OR INFERENCE PROCEDURE

**Offline training (simulation):** Standard PPO with domain randomisation. No modifications to the training procedure.

**Online adaptation (deployment):** At each step: (1) execute policy, collect transitions; (2) update transition scores from critic TD errors; (3) sample batch from priority queue; (4) apply one PPO + entropy-regularised critic update. Total compute per step equivalent to one standard PPO update.

## WHAT THE GUARANTEE SAYS

The entropy regularizer prevents critic collapse: H(V_phi(s)) >= H_min > 0 for all t, ensuring policy gradient variance stays non-zero. This provides a necessary condition for adaptation progress; convergence rate depends on the magnitude of distribution shift and data collection speed.

## EXPERIMENTAL FINDINGS

**Go2 (MuJoCo Playground, payload shift):** TEMPO reaches 80% of pre-shift performance within 5 minutes; standard PPO fine-tuning diverges.

**Go1, G1 (four shift types):** Consistent gains; average +42% over non-adapted baseline.

**Physical Unitree Go2:** TEMPO with 2 kg added payload recovers stable locomotion within 10 minutes; non-adapted policy fails to sustain a trot gait.

**Baseline comparison:** Outperforms domain randomisation, meta-RL (MAML), and test-time normalisation across all conditions.

## ABLATIONS AND INTERPRETATION

**Entropy regularizer only:** Without score-based memory, adaptation is slower (-25% performance at 5 min); both components are needed.

**Score-based memory only:** Without entropy regularization, the critic still diverges in 3 of 4 shift scenarios.

**Priority exponent alpha:** Performance robust to alpha in [0.4, 0.7]; alpha < 0.4 approaches uniform sampling and loses most gains.

**Batch size:** Adaptation improves up to batch size 64 and plateaus.

## REFERENCE

Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos. **Efficient Test-Time Adaptation For Robot Policies.** arXiv:2610.20318. NeurIPS 2026. https://arxiv.org/abs/2610.20318